

Monopoly

What is a monopoly?

- A *monopoly* refers to a firm that is the *only* producer of a good or service with no close substitutes.
- If a buyer or seller has the ability to noticeably affect market prices, that person/firm has *market power*.
- Competitive markets have so much competition that *no one* has the ability to affect market prices.
- Monopoly has very strong market power - it is the industry.

Characteristics of a monopoly

- A single producer – *no competition*
- Good or service has *no close substitutes*
- *Barriers to entry* into the market exist
- Firm is a price-maker or a *price-setter*

Real-life monopolies

- In reality, a company does not need to have 100% market share to operate like a monopoly
- A single company can have a huge share in the industry in which it operates, allowing it to *behave like a monopoly*
- e.g. Google as a search engine platform, Facebook as a social media platform

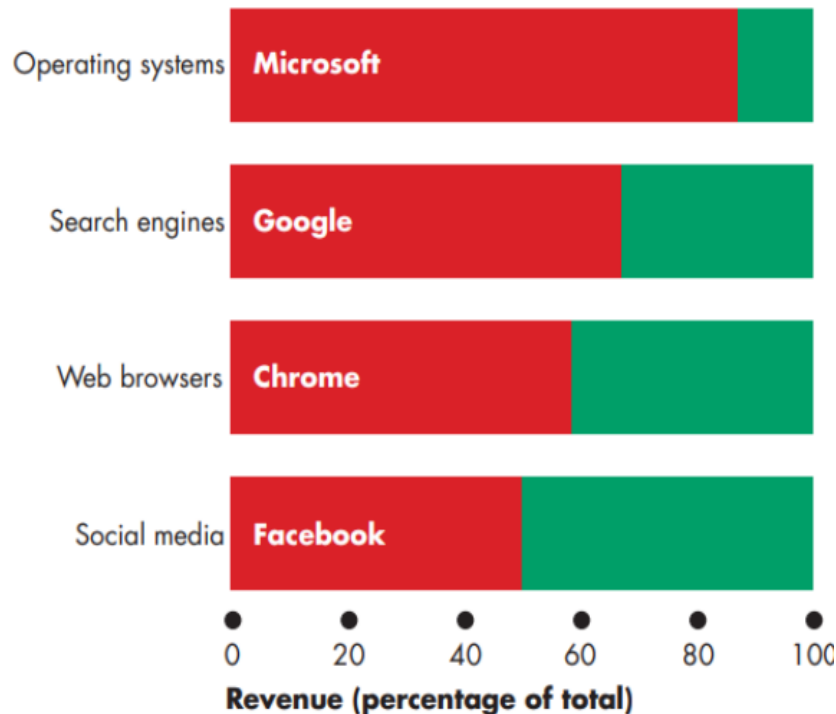


Figure 1 Market Shares

Barriers to entry

Two types of barriers to entry

1. *Natural barriers to entry*

- a *natural monopoly* is created in a market in which economies of scale enable one firm to supply the entire market at the lowest possible cost
- can arise due to high startup costs/fixed costs
- can arise due to *economies of scale* – a large output is needed to be produced to recover fixed costs
- Example: Dhaka WASA (costly to set up water supply system but marginal cost to supply an additional household is practically zero)

Barriers to entry

2. *Legal barriers to entry*

- some monopolies (*legal monopolies*) may be protected by *legal barriers* to entry such as government license, patent or a monopoly franchise
- ensures that firms are motivated to innovate and invest in *costly and time-consuming* research and development
- *Patents* give firms the exclusive right to produce and sell a particular product for a given time period
- Example: patent for a malaria or covid vaccines

Price setting strategies

- A monopolist is a price-setter. All monopolies face a trade-off between price and the quantity sold. To sell a larger quantity, the monopolist must charge a lower price.

It can use different price-setting strategies:

1. *Single Price Monopoly*

- A single-price monopoly is a monopoly that must sell each unit of its output for the same price to all its customers (e.g. DWASA charging same rate for water supply)

2. *Price-discriminating Monopoly*

- A price-discriminating monopoly charges different prices for different units of a good/service. Might charge different prices to different consumers as well (e.g. different age groups are often charged different prices for entry tickets at a museum/zoo, bus/air tickets during holiday season)

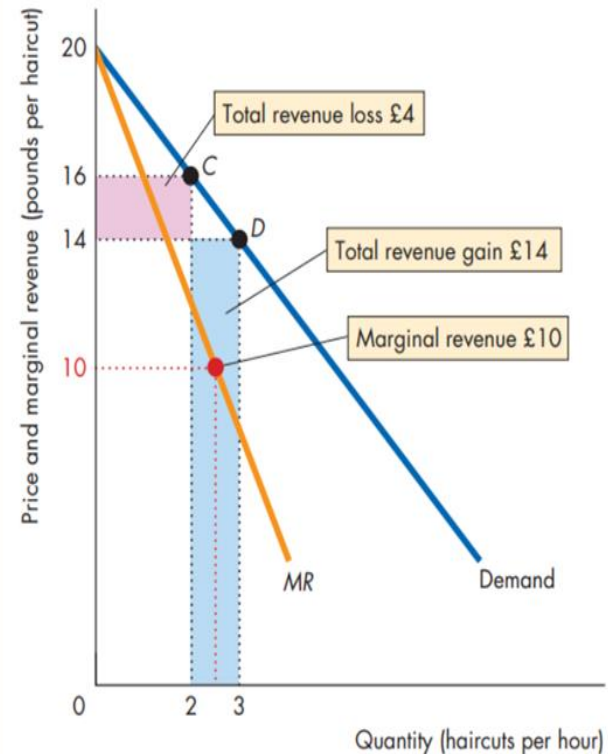
Single-price monopoly

- Recall, a perfectly competitive firm faces a perfectly elastic demand curve
- For a monopoly, the demand curve it faces is the same as the market demand curve (the monopoly is the entire market – no other producers)
- *Demand is usually downward sloping* – as you decrease price, quantity demanded increases (law of demand)
- A monopolist faces a trade-off between the price it charges and the quantity sold. In order to sell a larger quantity, the monopolist must charge a lower price.

Relationship between Price & Marginal Revenue

- Let's look at the demand curve for Gina's Cut and Dry, the only hairdressing salon within a 15 mile radius of a North Yorkshire town.
- In order to sell more haircuts per hour, Gina needs to lower her price
- Recall, **Total Revenue = Price X Quantity**
- **Marginal revenue** is the change in total revenue that results from a one unit increase in quantity sold
- Note that for each output level, marginal revenue is lower than the price (i.e. the MR curve lies below the demand curve)
- When the price is lowered to sell one more unit, **two opposing forces** affect total revenue.
- The lower price results in a revenue loss and the increased quantity sold results in a revenue gain.

Figure 12.2 Demand and Marginal Revenue



- For example, at a price of £16, Gina sells 2 haircuts (point C).
- If she lowers the price to £14, she sells 3 haircuts (point D)
- This means she has a revenue gain of £14 on the third haircut which is equal to the new price.
- But she now receives only £14 on the first two as well – £2 less than before.
- She loses £4 of revenue on the first 2 haircuts.
- To calculate marginal revenue, she must deduct this amount from the revenue gain of £14.
- So her marginal revenue is £10, which is less than the price (£14).

	Price (<i>P</i>) pounds per haircut)	Quantity demanded (<i>Q</i>) (haircuts per hour)	Total revenue ($TR = P \times Q$) (pounds)	Marginal revenue ($MR = \Delta TR / \Delta Q$) (pounds per haircut)
A	20	0	0	
				 18
B	18	1	18	
				 14
C	16	2	32	
				 10
D	14	3	42	
				 6
E	12	4	48	
				 2
F	10	5	50	

Marginal revenue and Elasticity

Why do firms care about Price Elasticity of Demand?

- A rise in price causes quantity demanded to decrease (law of demand)

Does a rise in price cause total revenue earned by a firm to increase or vice versa?

- *Marginal Revenue = change in TR / change in quantity sold*
- *Total Revenue = Price X Quantity Sold*
- Change in revenue depends on the *elasticity of demand*

Recall from Elasticity: Total Revenue and Elasticity

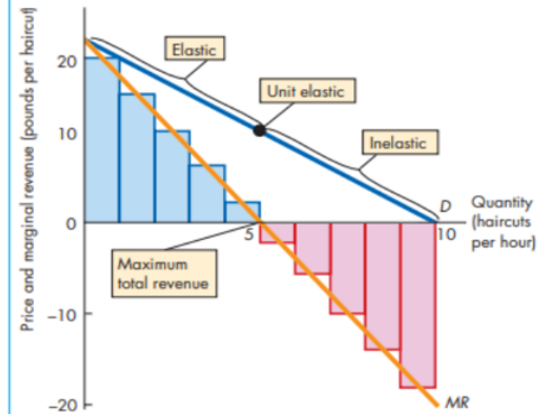
- If demand is elastic, a 1 percent price cut increases the quantity sold by more than 1 percent and *TR increases*.
- If demand is inelastic, a 1 percent price cut increases the quantity sold by less than 1 percent and *TR decreases*.
- If demand is unit elastic, a 1 percent price cut increases the quantity sold by 1 percent and so *TR does not change*.

Marginal revenue and Elasticity

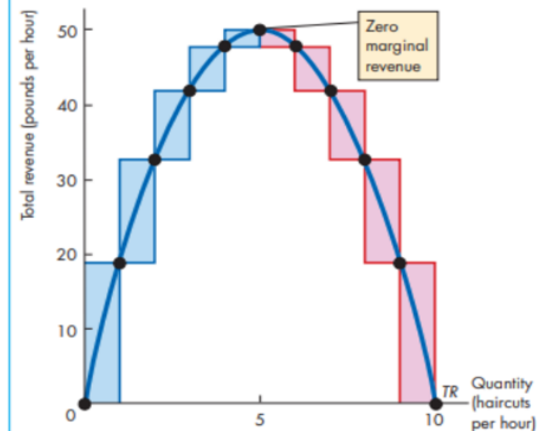
- If demand is *elastic*, a fall in price brings an *increase in total revenue* – the increase in revenue from the increase in quantity sold outweighs the decrease in revenue from the lower price – and *marginal revenue is positive*.
- If demand is *inelastic*, a fall in the price brings a *decrease in total revenue* – the increase in revenue from the increase in quantity sold is outweighed by the decrease in revenue from the lower price – and *marginal revenue is negative*.
- If demand is *unit elastic*, *total revenue does not change* – the increase in revenue from the increase in quantity sold offsets the decrease in revenue from the lower price – and *marginal revenue is zero*.

Marginal Revenue & Elasticity

- Over the range of 0 to 5 haircuts, the demand curve is elastic – a price cut increases TR – **MR is positive**
- Over the range of 5 to 10 haircuts, the demand curve is inelastic – a price cut decreases TR – **MR is negative**
- At 5 haircuts an hour, the demand curve is unit elastic – total revenue reaches its peak (maximum point) and **MR is zero** at this point
- This relationship implies that a profit-maximizing monopoly never produces an output in the inelastic range of its demand curve.



(a) Demand and marginal revenue curves



(b) Total revenue curve

Profit-Maximization for a Monopoly

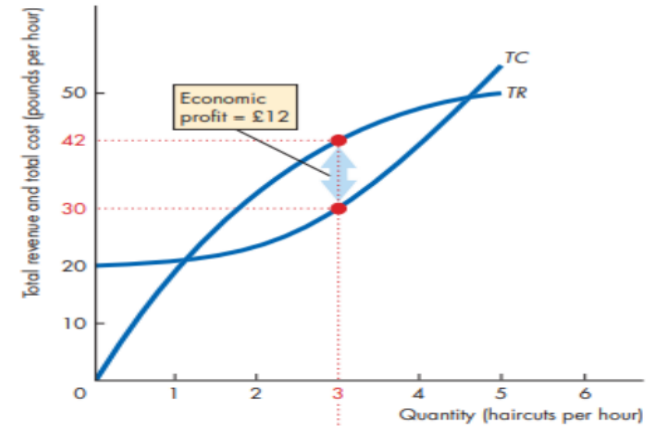
Table 12.1

A Monopoly's Output and Price Decision

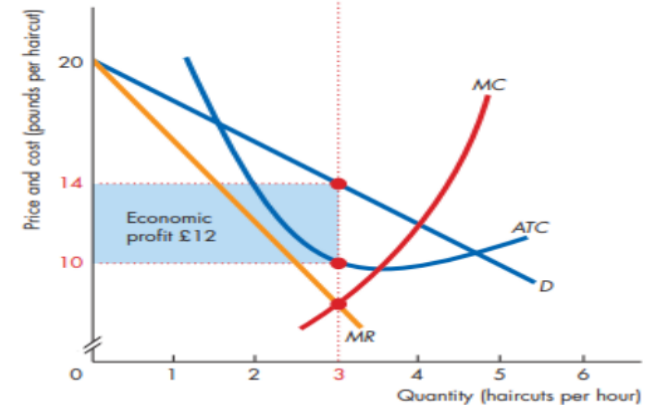
Price (P) (pounds per haircut)	Quantity demanded (Q) (haircuts per hour)	Total revenue ($TR = P \times Q$) (pounds)	Marginal revenue ($MR = \Delta TR / \Delta Q$) (pounds per haircut)	Total cost (TC) (pounds)	Marginal cost ($MC = \Delta TC / \Delta Q$) (pounds per haircut)	Profit ($TR - TC$) (pounds)
20	0	0		20		-20
			18		1	
18	1	18		21		-3
			14		3	
16	2	32		24		8
			10		6	
14	3	42		30		+12
			6		10	
12	4	48		40		+8
			2		15	
10	5	50		55		-5

- A monopoly will set its price and output at a level that maximizes its profits
- Profit = $TR - TC$
- Cost functions are same as we saw for a perfectly competitive firm
- We have seen how revenue, MR differs for a monopolist

- We can see from Panel A that TR increases at a decreasing rate while TC increases at an increasing rate
- Economic profits are maximized when the firm produces 3 haircuts an hour
- We know all firms maximize profits at the output where $MR = MC$
- For a monopoly, price exceeds MR (MR lies below the demand curve). This implies price exceeds MC as well.

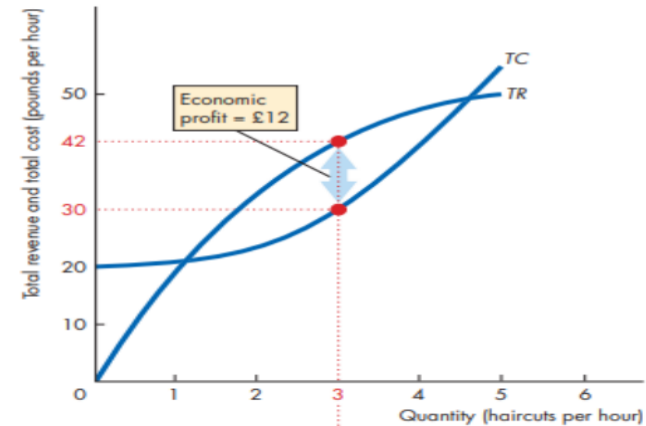


(a) Total revenue and total cost curves

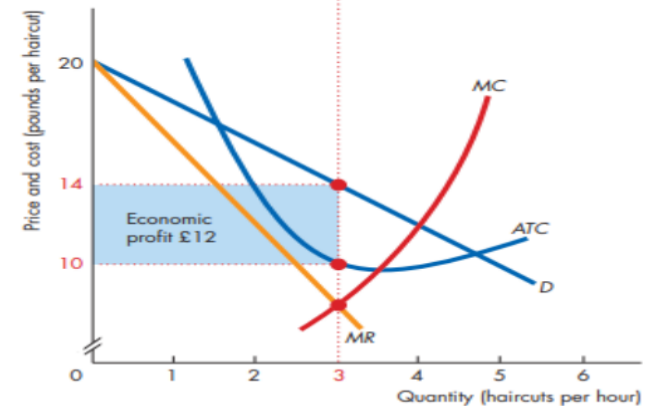


(b) Demand and marginal revenue and cost curves

- From the graph, we can see that at the profit maximizing level of output (3 haircuts), monopolist is charging a price = £14
- However, at this output level, ATC = £10
- So for each unit produced, profit = £4
- *Total profits = 3 X £4 = £12* (blue shaded region)
- If firms in a perfectly competitive market make a positive economic profit, new firms enter.
- In a monopoly, *barriers to entry prevent new firms from entering a market.*
- So a monopoly can make a positive economic profit and might continue to do so indefinitely.



(a) Total revenue and total cost curves



(b) Demand and marginal revenue and cost curves

Single-price Monopoly VS Perfect Competition

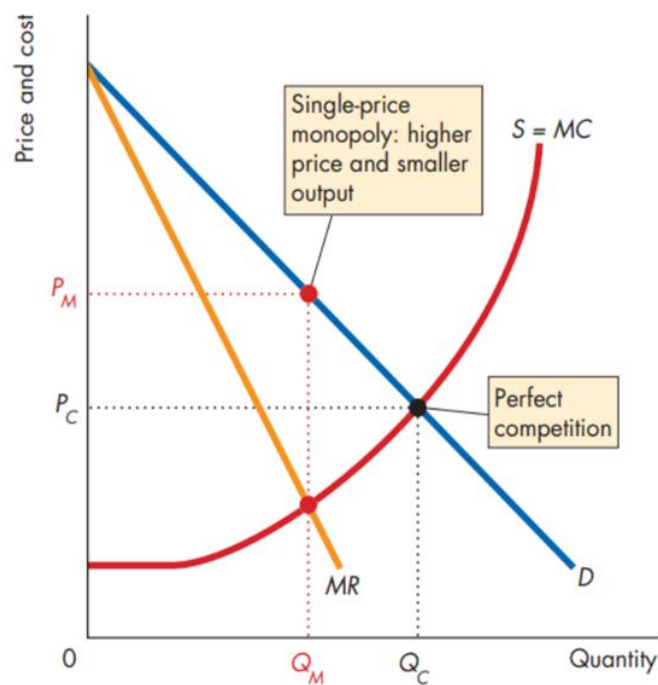
Perfect Competition

- The demand curve represents the overall market demand for the good. The supply curve represents the total supply in the market (supply across all firms combined)
- Under perfect competition, the quantity produced is Q_c and the price is P_c .

Monopoly

- In this case, since the monopoly is the only firm in the industry, the monopoly's MC curve is the same as the market supply curve.
- The monopoly maximizes profit by producing the quantity at which marginal revenue equals marginal cost.
- Under monopoly, the quantity produced is Q_M and the price is P_M .

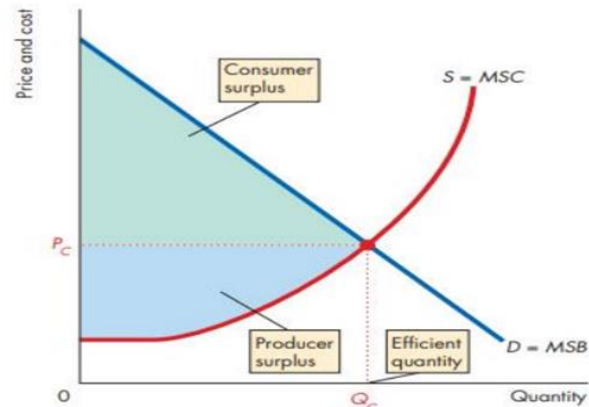
Compared with a perfectly competitive market, a single-price monopoly produces a smaller output and charges a higher price.



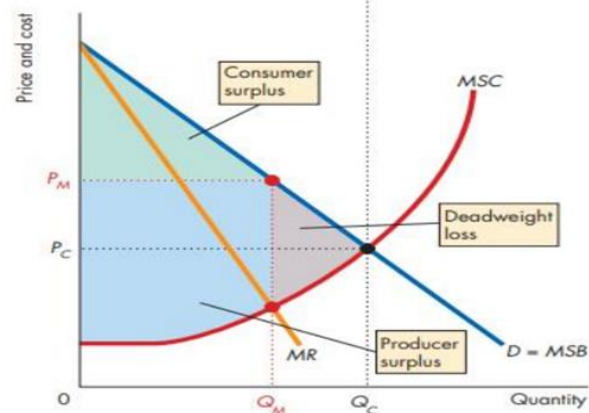
Single-price Monopoly VS Perfect Competition

Efficiency Comparisons

- Panel A shows that the output produced by a perfectly competitive firm is efficient.
- Consumer surplus is represented by the green triangle and Producer surplus is represented by the blue triangle
- Under perfect competition, total surplus is maximized
- A monopoly charges a higher price and restricts output at Q_M . This leads to a **deadweight loss (grey triangle)**
- Consumer surplus shrinks. A portion of this loss to consumers is a gain for the monopoly and increases the monopoly's producer surplus.
- Consumers also lose by getting less of the good, and this loss is part of the deadweight loss created by monopoly.
- Although the monopoly gains from a higher price, it loses some producer surplus because it produces a smaller output. That loss is another part of the deadweight loss created by monopoly.



(a) Perfect competition



(b) Monopoly

Welfare loss: Monopoly

- Monopoly power benefits monopolists but causes social welfare losses.
- In a competitive market, the equilibrium price and quantity maximize total surplus.
- Monopolists produce at a lower quantity than the efficient level.
 - Total surplus is not maximized.
 - Producer surplus (monopolist profit) increases.
 - Consumer surplus decreases.
 - The loss of total surplus is a deadweight loss (equal to the total surplus under perfect competition minus the total surplus under a monopoly).

Summary

- *Key concepts:*
 - Monopoly
 - Single-price monopoly
 - Profit-maximization of monopoly
 - Monopoly vs Perfect Competition
- *Reading: Chapter 12 of “Economics” by Parkin, Powell, Mathews 6th Ed.*